

Yibin (Leon) Liu

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Education

- School of Computing, National University of Singapore** [🌐] Jan 2027 – Jun 2031
(Expected)
Incoming Ph.D. Student in Computer Science
- **Advisor:** Prof. Weiyu Liu [🌐]
- Northeastern University, Shenyang, China** [🌐] Sep 2022 – Jun 2026
B.Eng. in Artificial Intelligence
- **Academic Average:** 88/100 overall; 93/100 in the junior year

About

My long-term goal is to build agentic robots that learn from real-world interaction and transfer knowledge reliably across new tasks, objects, and environments. My research connects **lifelong and compositional robot learning**, **uncertainty-aware multimodal reasoning and planning**, and **grounded program synthesis for embodied agents**. I possess **full-stack research and engineering capabilities** in embodied AI for mobile manipulation, spanning simulation development, model training, real-world robot deployment, and benchmark optimization.

Publications

^{*}Equal contribution. [†]Corresponding author.

- LingBot-VLA 2.0: From Foundation to Application, Improving VLA Models in Practice** [🌐]
Technical Report, Robbyant, 2026 **Core Contributor**
Robbyant Team. Contributed mobile-manipulation capabilities to LingBot-VLA.
- From Passive Observer to Active Critic: Reinforcement Learning Elicits Process Reasoning for Robotic Manipulation** [🌐]
ECCV 2026 **First Author**
Yibin Liu^{*}, Yaxing Lyu^{*}, Daqi Gao^{*}, Zhixuan Liang, Weiliang Tang, Shilong Mu, Xiaokang Yang, Yao Mu[†]
- RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation** [🌐]
ICML 2026 **Co-first Author**
Tianxing Chen^{*}, Zhanxin Chen^{*}, Baijun Chen^{*}, Zijian Cai^{*}, **Yibin Liu**^{*}, Zixuan Li^{*}, et al., Ping Luo[†], Yao Mu[†]
A scalable bimanual-manipulation benchmark spanning five embodiments, 50 tasks, and more than 700 assets.
- R3DP: Real-Time 3D-Aware Policy for Embodied Manipulation** [🌐]
ECCV 2026 **Co-author**
Yuhao Zhang, Wanxi Dong, Yue Shi, Yi Liang, J. Gao, Q. Yang, Yaxing Lyu, Zhixuan Liang, **Yibin Liu**, et al., Xiaokang Yang, Yanyan Xu, Yao Mu
- Vlaser: Vision-Language-Action Model with Synergistic Embodied Reasoning** [🌐]
ICLR 2026 **Co-author**
Ganlin Yang, Tianyi Zhang, Haoran Hao, Weiyun Wang, **Yibin Liu**, Dehui Wang, Guanzhou Chen, Zijian Cai, Juntong Chen, Weijie Su, Wengang Zhou, Yu Qiao, Jifeng Dai, Jiangmiao Pang, Geng Zhang[†]
- The Homework Wars: Exploring Emotions, Behaviours, and Conflicts in Parent-Child Homework Interactions** [🌐]
ACM IMWUT / UbiComp 2025 **Co-author**
Nan Gao^{*}, **Yibin Liu**, Xin Tang, Yanyan Liu, Chun Yu, Yun Huang, Xuhai Xu, Jun Wei, Yuanchun Shi
- HyCodePolicy: Hybrid Language Controllers for Multimodal Monitoring and Decision in Embodied Agents** [🌐]
ICCV 2025 Workshop on Multi-Modal Reasoning for Agentic Intelligence **First Author**
Yibin Liu^{*}, Zhixuan Liang^{*}, Zhanxin Chen^{*}, Tianxing Chen, Mengkang Hu, Wanxi Dong, Congsheng Xu, Zhaoming Han, Yusen Qin, Yao Mu[†]

Research Experience

University of North Carolina at Chapel Hill

Jun 2025 – Mar 2026

Research Intern; Advisor: Prof. Mingyu Ding[✉]

Remote

- Developed spatial-understanding and action-reasoning methods for multimodal language models in robotic manipulation using language instruction and AR-guided interaction.

Shanghai Jiao Tong University

Mar 2025 – Jan 2026

Research Assistant; Advisor: Prof. Yao (Mark) Mu[✉]

Shanghai, China

- Led robot-policy code generation for scalable bimanual manipulation as a core contributor to RoboTwin 2.0.

Pervasive HCI Lab, Tsinghua University[✉]

Jun 2024 – Jan 2025

Research Assistant; Advisors: Prof. Nan Gao[✉], Prof. Chun Yu[✉]

Beijing, China

- Applied LLM-assisted qualitative analysis to infer human behavior and emotional states from parent-child dialogue data.

Industry Experiences

Joy Future Academy, JD.COM[✉]

Aug 2026 – Present

Tech Genius Team (TGT) Intern

Shenzhen, China

- Developing compositional generation methods for long-horizon and mobile manipulation.

Robbyant[✉] (part of Ant Group[✉])

Mar 2026 – Aug 2026

Research Intern

Shanghai, China

- Built mobile-manipulation foundation models across VLA pre-training, post-training, real-robot deployment, and automated benchmark verification.

Horizon Robotics / D-Robotics

Jun 2025 – Jan 2026

Robotics Agent Intern

Beijing, China

- Developed RDK Copilot, an LLM-powered assistant for robotics coding, environment setup, testing, and documentation.

Academic Activities

- **Community:** Co-Founder of VapourX[✉], an open community for embodied AI beginners, enthusiasts, and researchers.
- **Workshop Service:** Student Committee, TriFusion Workshop at SIGGRAPH Asia 2025.
- **Academic Service:** Reviewer for CHI 2025 and Chinese CHI 2024.

Awards

- 2025.10 **Best Poster Award**, IROS 2025 RoDGE Workshop (RoboTwin 2.0).
- 2025.10 **Best Paper Finalist**, IROS 2025 RoboGen Workshop (RoboTwin 2.0).
- 2024.11 Outstanding Individual in Technological Innovation, Northeastern University.
- 2024.05 **Finalist**, Mathematical Contest in Modeling (MCM/ICM 2024; **top 1.69% of 10,387 teams**).
- 2023.10 National Third Prize, RoboCup China Competition Simulation 3D League.
- 2023.10 National Second Prize, FIRA SimuroSot China Competition.
- 2023.11 Future Technology Taihu Scholarship and Excellent Student Scholarship, Northeastern University.